

Approximate Continuity Restores the Necessary Obstruction for Deep Complex-Valued Networks

Run fa_banach_001 — lane 0

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Status: candidate partial result, likely valid. Everywhere approximate continuity is a natural hypothesis strictly weaker than continuity under which Voigtlaender’s necessary condition becomes valid. Together with his sufficient theorem, it gives a complete characterization inside this larger class. No claim of optimality of the hypothesis is made.

1 Source problem

Felix Voigtlaender’s *The universal approximation theorem for complex-valued neural networks*, arXiv:2012.03351, studies fixed-depth, arbitrarily wide complex-valued feedforward networks [1]. Let $\mathcal{NN}_{\sigma,L}^d$ denote the resulting scalar-output network class on $\mathbb{C}^d$ with activation $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ and L hidden layers.

The source class $\mathcal{M}$ consists of the locally bounded activations for which the closure of the set of ordinary discontinuity points has planar Lebesgue measure zero. Source Theorem 1.4 proves, for $L \geq 2$, that $\mathcal{NN}_{\sigma,L}^d$ is universal whenever $\sigma \in \mathcal{M}$ satisfies neither of the following:

- (a) $\sigma(z) = p(z, \bar{z})$ almost everywhere for some $p \in \mathbb{C}[X, Y]$;
- (b) $\sigma = g$ almost everywhere or $\sigma = \bar{g}$ almost everywhere for an entire function $g : \mathbb{C} \rightarrow \mathbb{C}$.

Its converse assumes that σ is continuous. The remark immediately after the theorem, on arXiv PDF page 4, asks for natural hypotheses weaker than continuity under which the necessary condition still holds.

2 Candidate result

Write λ for planar Lebesgue measure. A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is *approximately continuous at z_0* if, for every $\varepsilon > 0$,

$$\lim_{r \downarrow 0} \frac{\lambda(\{z \in B_r(z_0) : |f(z) - f(z_0)| < \varepsilon\})}{\lambda(B_r(z_0))} = 1.$$

Thus ordinary neighborhoods are replaced by sets of density one.

Theorem 1 (complete criterion under approximate continuity). *Let $\sigma \in \mathcal{M}$ be approximately continuous at every point of $\mathbb{C}$. For every $d \in \mathbb{N}$ and $L \geq 2$, the class $\mathcal{NN}_{\sigma,L}^d$ has the universal approximation property if and only if neither (a) nor (b) above holds.*

If either obstruction holds, the stronger source conclusion remains true: there are $F \in C_c(\mathbb{C}^d; \mathbb{R})$ and $\varepsilon > 0$ such that

$$\|F - \Phi\|_{L^1(B_1(0))} \geq \varepsilon \quad \text{for every } \Phi \in \mathcal{NN}_{\sigma,L}^d.$$

example, it is well-known (see for instance [16, Remark 2]) that bounded polyharmonic functions are necessarily constant. Thus, the theorem shows that if $\sigma \in \mathcal{M}$ is bounded but not equal to a constant (almost everywhere), then $\mathcal{NN}_\sigma^d$ is universal.

Furthermore, the theorem shows that any “bad” activation function σ necessarily has to be smooth (possibly after changing it on a null-set). Thus, any activation function that can not be made smooth by changing it on a null-set is a “universal” activation function.

Let us denote by $\mathcal{NN}_{\sigma,L}^d$ the set of complex-valued feedforward neural networks with activation function σ , input dimension d , and L hidden layers; a precise definition is given in Definition 4.9 below. The following theorem characterizes the activation functions σ for which the deep network class $\mathcal{NN}_{\sigma,L}^d$ ($L \geq 2$) is universal. Contrary to the real setting, there are strictly more “universal” activation functions for deep networks than for shallow ones.

Theorem 1.4. *Let $\sigma \in \mathcal{M}$ and assume that none of the following properties hold:*

- a) *we have $\sigma(z) = p(z, \bar{z})$ for almost all $z \in \mathbb{C}$, where $p \in \mathbb{C}[X, Y]$ is a complex polynomial of two variables,*
- b) *we have $\sigma = g$ almost everywhere or $\sigma = \bar{g}$ almost everywhere, where $g : \mathbb{C} \rightarrow \mathbb{C}$ is an entire holomorphic function.*

Then, for each $L \in \mathbb{N}_{\geq 2}$ and each $d \in \mathbb{N}$, the class $\mathcal{NN}_{\sigma,L}^d$ of deep complex-valued neural networks with activation function σ and L hidden layers has the universal approximation property.

Conversely, if $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ is continuous and satisfies a) or b), then $\mathcal{NN}_{\sigma,L}^d$ does not satisfy the universal approximation property for any $d, L \in \mathbb{N}$.

Remark. Note that the necessary condition requires σ to be continuous, whereas for the sufficient condition it is only assumed that $\sigma \in \mathcal{M}$. The additional continuity assumption is not simply a proof artifact. In fact, we will see in Example 4.13 that there does exist a discontinuous activation function $\sigma \in \mathcal{M}$ that coincides almost everywhere with a polynomial $p(z, \bar{z})$, but such that $\mathcal{NN}_{\sigma,L}^d$ is nevertheless universal for all $L \in \mathbb{N}_{\geq 2}$ (but not for $L = 1$). We leave it as future work to determine natural conditions on σ that are weaker than continuity, but under which a necessary condition as in the above theorem still holds.

We would like to emphasize that any *discontinuous* function σ satisfying (at least) one of the two conditions in Theorem 1.4 has to be quite pathological: Such a function is discontinuous, but coincides almost everywhere with a continuous function. In that case, it is usually much more natural to work with the “continuous version” of σ instead of with σ itself. Moreover, such a function cannot be evaluated numerically in a reliable way and is therefore useless for practical applications.

Figure 1: Source Theorem 1.4 and its future-work question, arXiv PDF page 4.

Proof intuition

The exceptional values of an almost-everywhere representative matter because deep networks compose the activation with itself. Approximate continuity removes precisely this freedom: a density-one approach to every point sees the continuous representative, so the value at that point is forced to equal the representative’s value. Consequently, an activation which is almost everywhere polynomial, entire, or conjugate-entire is pointwise so. The source obstruction proof then applies verbatim. Its sufficient theorem already requires only $\sigma \in \mathcal{M}$, so the two directions meet.

Lemma 2 (density-one rigidity). *Let $f, h : \mathbb{C} \rightarrow \mathbb{C}$. Suppose that $f = h$ almost everywhere, f is approximately continuous at z_0 , and h is continuous at z_0 . Then $f(z_0) = h(z_0)$.*

Proof. Fix $\eta > 0$. Approximate continuity says that

$$A_r := \{z \in B_r(z_0) : |f(z) - f(z_0)| < \eta\}$$

has positive measure for every sufficiently small r . Continuity of h allows r to be chosen so that $|h(z) - h(z_0)| < \eta$ throughout $B_r(z_0)$. Since the set on which $f = h$ has full measure, there is a point $z \in A_r$ with $f(z) = h(z)$. Hence

$$|f(z_0) - h(z_0)| \leq |f(z_0) - f(z)| + |h(z) - h(z_0)| < 2\eta.$$

Letting $\eta \downarrow 0$ proves the claim. $\square$

Proof of Theorem 1. Suppose first that (a) holds. The function $h(z) = p(z, \bar{z})$ is continuous, so Lemma 2, applied at every $z_0 \in \mathbb{C}$, gives $\sigma = h$ pointwise. In case (b), the same argument applies to $h = g$ or $h = \bar{g}$. In every case σ is therefore continuous and satisfies the corresponding pointwise obstruction. Voigtlaender's Theorem 4.12 now supplies the displayed L^1 lower bound, and in particular rules out universality [1].

Conversely, if neither (a) nor (b) holds, source Theorem 4.11 (equivalently, the sufficient direction of Theorem 1.4) applies directly to $\sigma \in \mathcal{M}$ and yields universality for every $d \in \mathbb{N}$ and $L \geq 2$. $\square$

3 The hypothesis is strictly weaker than continuity

The result is not merely continuity under a different name, even after intersecting with the source class $\mathcal{M}$.

Proposition 3. *There is a bounded $\sigma \in \mathcal{M}$ which is approximately continuous at every point of $\mathbb{C}$ but is not continuous.*

Proof. Identify the positive real axis with a subset of $\mathbb{C}$. Put

$$a_n = 2^{-n}, \quad r_n = 2^{-4n-4}, \quad D_n = B_{r_n}(a_n) \quad (n \geq 1).$$

The closed disks $\bar{D}_n$ are pairwise disjoint. Let $\psi(w) = \max\{1 - |w|, 0\}$ and define

$$\sigma(z) = \sum_{n=1}^{\infty} \psi\left(\frac{z - a_n}{r_n}\right).$$

At most one summand is nonzero at each point. Away from the origin the disk family is locally finite, and each bump vanishes on its boundary, so σ is continuous on $\mathbb{C} \setminus \{0\}$. But $\sigma(0) = 0$ while $\sigma(a_n) = 1$ and $a_n \rightarrow 0$; hence σ is discontinuous at 0.

It remains to check approximate continuity at 0. Let $E = \bigcup_n D_n$, and choose N with $2^{-(N+1)} < R \leq 2^{-N}$. No D_n with $n < N$ meets $B_R(0)$, and therefore

$$\frac{\lambda(E \cap B_R(0))}{\lambda(B_R(0))} \leq \frac{\sum_{n \geq N} r_n^2}{R^2} \leq \frac{2^{-6N-6}}{1 - 2^{-8}} \rightarrow 0.$$

Since $\sigma = 0$ off E , every set $\{z : |\sigma(z) - \sigma(0)| < \varepsilon\}$ has density one at 0. Thus σ is approximately continuous there, and it is ordinarily continuous (hence approximately continuous) everywhere else. Finally, $0 \leq \sigma \leq 1$ and its discontinuity set is exactly $\{0\}$, so $\sigma \in \mathcal{M}$. $\square$

Remark 4. The source's pathological universal activation (Example 4.13) fails the new hypothesis in the expected place: on the negative real axis its prescribed point value differs from the approximate limit supplied by the values off the axis. Approximate continuity therefore excludes that null-set coding mechanism without imposing ordinary continuity everywhere.

4 Verification, limitations, and novelty check

The proof has two dependencies: the elementary density-one rigidity lemma proved above and Voigtlaender's published Theorems 4.11–4.12. No numerical or symbolic computation is used. The main points for human verification are the all-points quantifier in approximate continuity, the

pointwise upgrade in Lemma 2, and the fact that the source’s L^1 obstruction can be invoked after that upgrade.

This answers the future-work direction by giving a natural strict weakening and a complete characterization within it. It does not characterize all discontinuous representatives for which non-universality holds, and it does not assert that approximate continuity is minimal.

A bounded novelty search through 9 August 2026 checked the run indexes; the exact phrase “weaker than continuity” from arXiv:2012.03351; arXiv/web queries combining complex-valued networks with approximate continuity; and the nearby later papers arXiv:2303.16813 and arXiv:2305.16910, which cite the source but work with smooth or continuous activations [2, 3]. No direct statement of the extension above was found. Mathematical confidence is high; novelty confidence is moderate because the rigidity step is elementary and could be implicit in the literature.

References

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